

Project Phase 1: Lexical Analyzer for Tiny Language

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User Interface (UI):

The screenshot displays a Windows-style application window titled "Form1". On the left is a text area containing C-like code:

```
int sum(int a, int b)
{
    return a + b;
}

int main()
{
    int x,y;
    read x;
    y := sum(x,5);
    write y;
    return 0;
}
```

To the right of the code editor is a table with two columns: "Lexeme" and "Token". The table contains the following data:

Lexeme	Token
int	Keyword
sum	Identifier
(	Left_Paren
int	Keyword
a	Identifier
,	Comma
int	Keyword
b	Identifier
)	Right_Paren
{	Left_Brace
return	Keyword
a	Identifier
,	Comma

Below the table is a large, empty rectangular box. At the bottom center of the window is a button labeled "Scan Code".

Form1

```

int sum(int a, int b)
{
return a + b;
}

int main()
{
int x,y;
read x;
y := sum(x,5);
write y;
return 0;
}

```

Lexeme	Token
+	Plus_Op
b	Identifier
:	Semicolon
}	Right_Brace
int	Keyword
main()	Function_Call
{	Left_Brace
int	Keyword
x	Identifier
,	Comma
y	Identifier
:	Semicolon
read	Keyword

Scan Code

Form1

```

int sum(int a, int b)
{
return a + b;
}

int main()
{
int x,y;
read x;
y := sum(x,5);
write y;
return 0;
}

```

Lexeme	Token
read	Keyword
x	Identifier
:	Semicolon
y	Identifier
:=	Assignment_Op
sum	Identifier
(	Left_Paren
x	Identifier
,	Comma
5	Number
)	Right_Paren
:	Semicolon
write	Keyword

Scan Code

Form1

```
int sum(int a, int b)
{
    return a + b;
}

int main()
{
    int x,y;
    read x;
    y := sum(x,5);
    write y;
    return 0;
}
```

	Lexeme	Token
	,	Comma
	5	Number
	)	Right_Paren
	;	Semicolon
	write	Keyword
	y	Identifier
	;	Semicolon
	return	Keyword
	0	Number
	;	Semicolon
	}	Right_Brace
*		

Scan Code

Git Hub Repository: <https://github.com/samymagdi/Tiny-Language-Lexer.git>